A rotating electric field

Can a rotating E field make a B field? Real fields, and the magnetic-charge case labelled hypothetical — CLAUDE.md §3f, register entry H2

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Summary

This is the inverse of the homopolar dynamo (*examples/homopolar*): there a conductor spinning in a magnetic field B gives a voltage; what does a *rotating electric field* E give? The example keeps two questions apart. First the real physics: a magnetic field comes from moving charge (Rowland 1876) and from a changing electric field (the displacement current of the Ampère–Maxwell law), checked with the quaternion field operators of milestone M9h. Then, opt-in and labelled, the hypothetical physics: the force a magnetic charge would feel in these fields (the dual pair of Joot's Theorem 3.16), kept out of the real momentum ledger.

- **A rotating E pattern by itself makes no B.** B comes from the charge that moves, or from ∂E/∂t where the field really changes in time, exactly as the Ampère–Maxwell law says.
- **Rowland disk** (20,000 rpm): $B_z = 1.316\times10^{-9}$ T at the centre and 4.496×10^{-10} T at z = 5 cm, equal to the analytic values.
- **Charged ring**: the same static E pattern at rest and spinning, but B at the centre is 0 T at rest and 1.26×10^{-8} T spinning (the mirror of the Faraday-disk paradox).
- **Röntgen–Eichenwald disk** (dielectric spinning in a fixed E): $B_r = -3.46\times10^{-10}$ T against the thin-disk -3.5×10^{-10} T. The $1/c^2$ is why the electric mirror of the homopolar effect is nanotesla where the homopolar one is volts.
- **Hypothetical magnetic charge** ($q_m = 1$ A·m, none ever observed): forces of order 10^{-9} N, booked only in the hypothetical column; the real net force stays 0.

What is being asked

Standard prediction. By the Ampère–Maxwell law (Maxwell's Treatise, Vol. II, Art. 619: 4π C = curl H in his quaternion notation, where the total current C includes the displacement current ∂D/∂t), any time-varying electric field produces a magnetic field, and moving charge produces one (Rowland). A field pattern that only appears to rotate while the charges stay still produces none.

Hypothesis (register entry H2: magnetic charge, dual-symmetric Maxwell). The dual pair of the Lorentz force would push a magnetic charge q_m with $q_m(B - v\times E/c^2)$. No magnetic monopole has been observed, so in normal mode $q_m = 0$ and the dual pair gives zero force. With *em.Hypothetical{MagneticCharge: true}* the example computes what such a charge would feel, labelled hypothetical and kept in a separate ledger column (CLAUDE.md §3f).

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The five real cases

case	set-up	result	expected / check
1a. Rowland disk	$\sigma = 10^{-5}$ C/m², R = 10 cm, 20,000 rpm; B_z on the axis by summing 400 current rings	1.316×10^{-9} T at z = 0; 4.496×10^{-10} T at z = 5 cm	analytic $(\mu_0\sigma\omega/2)[(R^2 + 2z^2)/\sqrt{(R^2 + z^2)} - 2|z|]$: 1.316×10^{-9}, 4.496×10^{-10} T
1b. Rotating uniform E	1 MV/m rotating at 1 MHz, as between quadrature-driven plate pairs	$\|B\| = 1.75\times10^{-6}$ T at r = 5 cm	$(\mu_0\varepsilon_0/2)\cdot E\cdot\omega\cdot r$; Ampère–Maxwell residual 3.3×10^{-10}, Faraday residual 4.2×10^{-7} (the neglected $(\omega r/c)^2$ term)
1c. Charged ring	1 μC on 5 cm, at rest and spinning at 1000 rev/s	B at centre 0 T at rest, 1.26×10^{-8} T spinning	the E pattern is identical and static in both; B follows the moving charge
1d. Rotating charge pair	±1 nC on a 10 cm rod at 1000 rev/s	$\|B\| = 8.99\times10^{-12}$ T on the axis 2.5 cm up	the E pattern really rotates; Ampère–Maxwell residual 1.3×10^{-4} of the displacement term (stencil truncation)
1e. Röntgen–Eichenwald	$\varepsilon_r = 3$ disk, R = 10 cm, 1 mm thick, 50 rev/s in a fixed 1 MV/m	$B_r = -3.46\times10^{-10}$ T at r = 5 cm (ring sum)	thin-disk $-(\varepsilon_r - 1)\cdot E\cdot v/c^2 = -3.5\times10^{-10}$ T

Case 1e: the electric mirror of the homopolar disk

A dielectric moving in an electric field is magnetized, M = P × v (Minkowski; Jackson §11.10). The ring sum uses only the bound face charge $\varepsilon_0(\varepsilon_r - 1)E$ carried round as two opposite surface currents and Biot–Savart's μ_0, so the $1/c^2 = \mu_0\varepsilon_0$ comes out by itself, not by hand. Its deficit against the thin-disk value falls linearly in h/R (h the thickness): the test checks < 1.5 % at h/R = 0.01 and that halving h halves it. That $1/c^2$ is why the mirror effect is nanotesla where the homopolar one is volts: 1 T at 100 m/s gives 100 V/m, while 1 MV/m at 100 m/s gives about 1 nT.

Answer

A rotating E pattern by itself makes no B. B comes from the charge that moves (1a, 1c, 1d, 1e) or from $\partial E/\partial t$ where the field really changes in time (1b, 1d), exactly as the Ampère–Maxwell law says; the quaternion residuals confirm the fields are solutions of Maxwell's equations.

Hypothetical: a magnetic charge in these fields

Opt-in with *em.Hypothetical{MagneticCharge: true}*, magnetic charge $q_m = 1$ A·m. No magnetic charge has ever been observed (CLAUDE.md §3f). These numbers go to the hypothetical column of *em.MomentumLedger* only.

situation	hypothetical force	real net force
at rest at the Rowland disk's centre	1.32×10^{-9} N ($q_m\cdot B$)	0 (the momentum ledger's real column stays empty)
moving at 100 m/s through 1 MV/m	1.11×10^{-9} N ($q_m\cdot v\times E/c^2$)	—

Validation

- Each case is tested against its analytic value (*main_test.go*).
- Field solutions are checked with the M9h quaternion operators: Ampère–Maxwell and Faraday residuals as listed in the table (3.3×10^{-10} and 4.2×10^{-7} for 1b; 1.3×10^{-4} of the displacement term for 1d, from stencil truncation).
- Röntgen–Eichenwald: deficit against the thin-disk value < 1.5 % at h/R = 0.01, halving with h.

Limits of the model

- Case 1b keeps the quasi-static field; the Faraday residual is the neglected $(\omega r/c)^2$ term.
- The magnetic-charge results are hypothetical (H2, untested): they show what a monopole would feel, not that one exists.

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Status and reproducing the results

The real cases are standard electromagnetism and agree with their analytic values. Register entry H2 (magnetic charge, dual-symmetric Maxwell) stays **untested**: this example only books what a magnetic charge would feel, in the hypothetical column.

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go run ./examples/rotating_efield
go test ./examples/rotating_efield
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Sources: H. A. Rowland, Am. J. Sci. 15, 30 (1878); W. C. Röntgen, Ann. Phys. 35, 264 (1888); A. Eichenwald, Ann. Phys. 11, 1 (1903); J. D. Jackson, *Classical Electrodynamics*, 3rd ed., §11.10; Panofsky & Phillips, *Classical Electricity and Magnetism*, §18-3; J. C. Maxwell, *Treatise*, Vol. II, Art. 619; P. Joot, *Geometric Algebra for Electrical Engineers*, Theorem 3.16 (reference only).

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